

POTPRESS · DESIGN DECISIONS REGISTER

PotPress design decisions register

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DOCUMENT CONTROL

Document ID	PPR-DEC-001	Version	0.2
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potpress		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened; open decisions moved out of the build plan

PotPress design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Pump slot width	30 mm (as built in the model) or back to 25 mm	30 mm, with the guard distance check completed	Guard panel and slot	PPR-DDR-004, Q2; PPR-DDR-003
2	Pump handle outside the guard	Treat the 158 mm the handle stands out as operating space, or redesign	Treat it as operating space	Floor space beside the press	PPR-DDR-004, Q3
3	Demolding by tipping the mold on its pins	Confirm with a partner factory's potters, or change the method	Confirm with potters	Carriage and rail extension	PPR-DDR-004, Q5
4	Working pressing force	5 to 10 t assumed; confirm with a partner factory	Confirm before any pressing test	Jack, frame and mold checks	PPR-DDR-001, item 9
5	Guard opening check to ISO 13857	Check the slot and mesh distances against the tables	Complete the check	Guard panels	PPR-DDR-003

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Buy the bottle jack first	Its base, release screw and pump socket set the locating blocks, the interlock coupling and the notch angle	PPR-DDR-004, Q4
2	Mold cavity finish that releases the liner cleanly	Sets the hand finishing of both molds	PPR-DDR-004

■ Value engineering

Value-engineering target: USD 1,060 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,143 (USD 83 over the target). Main cost drivers and savings worth trying:

- The press alone is USD 1,060 and the QC rack USD 83. Read against the press alone, the estimate matches the target.
- Making the design constructable added USD 92 (PPR-DDR-004): the steel base plate under the female mold, the thicker 450 mm mold flanges, dowels and bushes (items 8 and 9, USD 44), floor anchors, the bolted jack plate and heavier feet (item 1, USD 22), the adapter disc and nut box (item 10, USD 9) and interlock details (item 21, USD 7).

- The fixed guards, front gate and interlock added USD 67 earlier (PPR-DDR-003); they are a safety requirement, so savings are sought elsewhere.
- Savings worth trying: cheaper dowels and bushes, and plywood QC rack shelves cut from offcuts.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items and recommendations	Amish: "i accept all your recommendations, go with them across all repos."	PPR-DDR-001, PPR-DDR-002
2026-09-27	Guarded version with mesh guards and an interlocked front gate; budget USD 1,060	Amish	PPR-DDR-003
2026-09-30	Design for construction: open-topped male mold, dowel location, M16 joints with spacer tubes, captive lead screw nut, interlock design and other changes that make the press buildable	Amish: "i accept your recommended changes on design that are currently being sent across for my approval"	PPR-DDR-004